

Fares Mohammed Nabil Elsherif

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OBJECTIVES

Dedicated Communications & Information Engineering student at Zewail City with proven expertise in C++, ESP32 architectures, and PID control. Deeply passionate about hardware-software integration, having successfully engineered real-time IoT networks, automated sensor systems, and custom electronic circuits. Seeking an internship role strictly focused on Embedded Systems Engineering.

TECHNICAL SKILLS

Languages: C++ (OOP, STL), Python (GUI, scripting)

Machine Control: closed-loop & PID control, real-time systems

Automation control: Sensor-actuator integration, automated control, process automation

Circuit Design: Industrial circuit design, NE555, CD4017, I2C, amplifiers, voltage regulation

Embedded Systems: Arduino, ESP32, ESP32-CAM, MPU6050 IMU, ESP-NOW, Blynk IoT dashboard, edge-based signal detection

Tools & Languages: Git, GitHub, signal processing, documentation | Arabic (native), English (professional)

PROJECTS

[\(view all projects\)](#)

MAWZOON — Self-Balancing Robot [\(view project\)](#) | ESP32 · MPU6050 · L298N · PID Control · C++

- Designing a 2-wheeled self-balancing robot on ESP32, using a **6-axis MPU6050 IMU (I2C)** for real-time tilt sensing that feeds a closed-loop **PID balance controller**.
- Engineered the power architecture — **2S (7.4V) 18650 Li-ion pack** with a **470–1000µF decoupling capacitor** on the 5V rail — paired with an L298N dual H-bridge driving 2 DC geared motors via independent PWM channels.

BEITI-TECH — AI Smart Home System [\(view project\)](#) | ESP32 · ESP32-CAM · ESP-NOW · Blynk · C++

- Built a **dual-ESP32 system (main unit + ESP32-CAM) linked over ESP-NOW**; implemented ESP-DL face recognition that auto-unlocks the door for known residents and sends Telegram alerts for unrecognized visitors.
- Integrated 6 sensor/actuator modules (DHT11, PIR, MQ-2 gas, flame, servo lock, buzzer) into a Blynk dashboard for automated lighting/fan control and gas/fire safety alerts; presented as Team 15 at Zewail City's 3rd Innovation Exhibition.

AQUASMART — IoT Smart Fish Farm Monitoring System [\(view project\)](#) | ESP32 · Blynk · C++

- Engineered a real-time water-quality system on ESP32 integrating 5 sensors (**pH, ±0.5°C DS18B20** temperature, water level, YF-S201 flow, and a derived dissolved-oxygen reading) with SD-card CSV logging and a Blynk dashboard.
- Derived a novel temperature/flow-based dissolved-oxygen equation, cutting hardware cost to ~1,200 EGP and removing the need for a dedicated DO sensor; validated over a **7-day, 60L trial (pH 7.0–8.2, DO 4.8–6.2 mg/L)** with automatic pump control.

Electronic Circuit Design Portfolio | Industrial Circuit Design · Debug · Documentation

- Designed, built, and debugged 5 functional circuits (DC power supply, traffic light controller, diode logic gates, amplifier); resolved 3+ design faults through fault isolation; produced a full annotated documentation portfolio.

EDUCATION

Zewail City of Science and Technology — B.Sc. Communications & Information Engineering

2025 – 2029

Coursework: Calculus, Physics, OOP, Electronic Circuits, Engineering Design

El-Sadat STEM High School — STEM Track | 89%

Graduated 2025

UNIVERSITY ACTIVITIES & CERTIFICATES

ZC Open House | Event Organizer

Guided over **160+ attendees** and managed on-site logistics across **15+ booths** during the university open house event.

ZC Career Fair 2026 | Student Usher

- Streamlined the registration and check-in process for over **200+** student attendees, optimizing crowd flow and minimizing wait times.
- Guided students through the fair layout, facilitating smooth interactions between attendees and companies' booths.

ZC Ambassador Team | Team Member

- Represented the university as an official ambassador, providing guidance, answering inquiries, and fostering a welcoming environment for prospective **High School students** and guests in any **university events**.

IEEE Student Branch

- Participated in **5+ embedded systems workshops** as an active student member, collaborating in small teams of **4-6 members** to prototype and troubleshoot hardware-software solutions.